CRM Application for Banquet Hall Booking

Introduction

In today's digital age, the event management industry is increasingly adopting technology to improve efficiency and customer experience. One important area is banquet hall booking, which traditionally relied on manual methods such as phone calls, in-person visits, and paper records. These methods often led to issues like double bookings, poor communication, and difficulty in managing customer data. To overcome these challenges, a **Customer Relationship Management (CRM) Application for Banquet Hall Booking** provides a modern and efficient solution.

This application serves as a centralized platform that simplifies the entire booking process. Customers can easily search for available banquet halls, check real-time availability, compare prices, and make bookings online. This not only saves time but also offers a convenient and transparent experience. The system also allows users to customize their bookings by selecting additional services such as catering, decoration, and seating arrangements.

A key feature of this system is its CRM functionality, which helps manage customer information effectively. It stores details such as contact information, booking history, and preferences, enabling businesses to provide personalized services and improve customer satisfaction. Additionally, the system supports automated notifications like booking confirmations and payment reminders, ensuring smooth communication.

For administrators, the application offers tools to manage bookings, monitor schedules, and generate reports. It also includes secure payment integration and analytics to track business performance.

Overall, a CRM-based banquet hall booking system enhances operational efficiency, reduces manual errors, and improves customer relationships, making it an essential tool for modern event management businesses.

It helps:

- Customers book halls efficiently
- Managers track bookings and availability
- Businesses improve customer relationships and revenue

Objectives

The primary objective of a CRM Application for Banquet Hall Booking is to develop an efficient, user-friendly, and reliable system that simplifies the process of managing banquet hall reservations while enhancing customer relationships. The system aims to replace traditional manual methods with a digital platform that ensures accuracy, transparency, and convenience for both customers and administrators.

One of the key objectives is to **streamline the booking process** by enabling customers to search for available banquet halls, check real-time availability, and make reservations online without any hassle. This reduces time consumption and eliminates the need for physical visits or repeated communication.

Another important objective is to **centralize customer data management**. The system stores customer details, booking history, and preferences in a structured manner, allowing businesses to provide personalized services and maintain long-term relationships with clients.

The application also aims to **improve operational efficiency** by providing administrators with tools to manage bookings, schedules, pricing, and resources effectively. This helps in minimizing errors such as double bookings and ensures better coordination among staff members.

Ensuring **secure and convenient payment processing** is another objective. The system integrates online payment options, allowing customers to make transactions safely while maintaining accurate financial records.

Additionally, the application focuses on **enhancing communication** through automated notifications such as booking confirmations, reminders, and updates. This ensures that customers are well-informed at every stage of the booking process.

Finally, the system aims to **support business growth** by providing analytical reports and insights, helping managers make informed decisions, identify trends, and improve overall service quality.

- Simplify banquet hall booking process
- Centralize customer data and interaction history
- Improve scheduling and resource allocation
- Automate billing and notifications
- Enhance customer experience

Scope of the System

The scope of the CRM Application for Banquet Hall Booking covers the development and implementation of a comprehensive system designed to manage all aspects of banquet hall reservations and customer relationships. The system is intended to serve as a centralized platform for customers, administrators, and staff, ensuring smooth coordination and efficient operations.

The application allows customers to browse available banquet halls, check real-time availability, view pricing and facilities, and make bookings online. It also enables users to customize their event requirements, such as catering, decoration, and seating arrangements. Customers can manage their bookings, track payment status, and receive notifications regarding confirmations and updates.

From the administrative perspective, the system provides tools to manage hall details, schedules, pricing, and booking approvals. Administrators can monitor all reservations, avoid scheduling conflicts, and maintain accurate records of transactions and customer data. The CRM functionality helps store and analyse customer information, including booking history and preferences, to improve service quality and customer engagement.

However, the scope is limited to banquet hall booking and related services within the organization. It does not include external vendor management or advanced event planning services unless integrated in future enhancements.

Overall, the system aims to provide a scalable, secure, and user-friendly solution that enhances efficiency, reduces manual effort, and improves customer satisfaction in banquet hall management.

The system is designed for:

- Banquet hall owners
- Event managers
- Customers (individuals or organizations)

It supports:

- Online booking
- Event scheduling
- Customer management
- Payment tracking
- Reporting & analytics

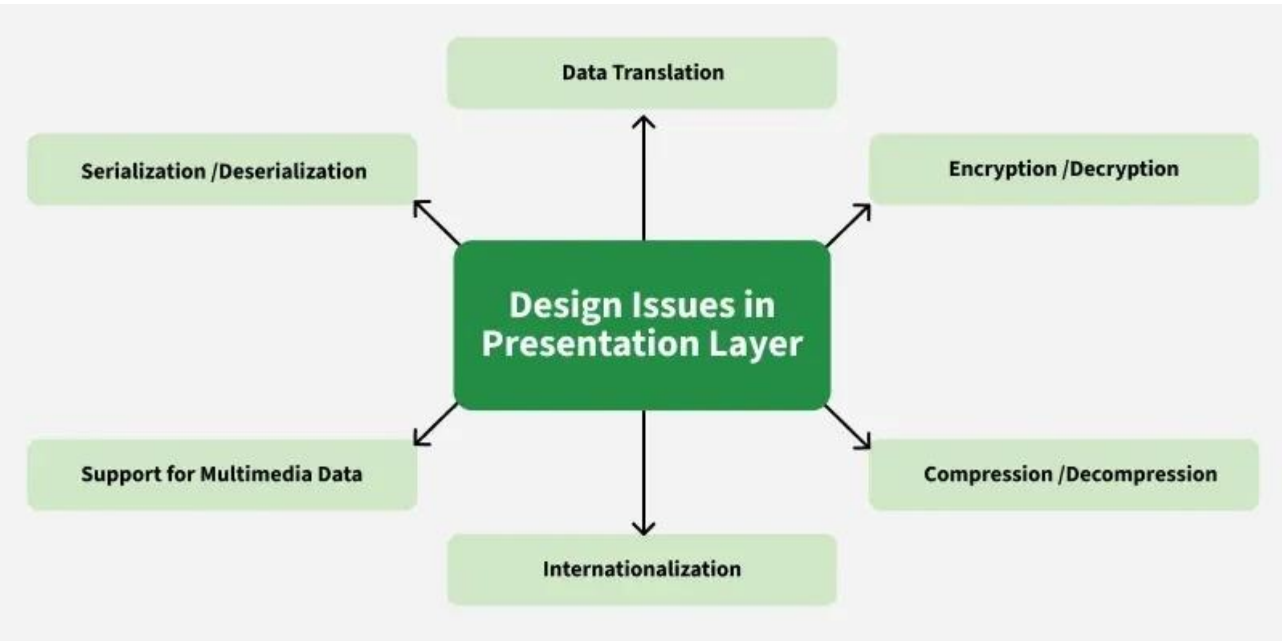
System Architecture

The system architecture of a **CRM Application for Banquet Hall Booking** is designed to provide a robust, scalable, and secure framework that connects customers, administrators, and staff while managing all operations related to banquet hall reservations and customer relationships. The architecture follows a **multi-tier client-server model**, typically a **three-tier structure**, comprising the **presentation layer (frontend)**, the **application layer (backend)**, and the **data layer (database)**. Additional modules such as authentication, reporting, notification systems, and API integration are also incorporated to enhance functionality.

1. Presentation Layer (Frontend)

The presentation layer serves as the interface between users and the system. It is responsible for all interactions with customers, administrators, and staff. Customers can explore available banquet halls, filter options based on capacity, pricing, and amenities, view hall details, and check real-time availability. They can also customize their bookings by selecting additional services such as catering, decoration, seating arrangements, and audio-visual equipment.

Administrators and staff access this layer through dedicated dashboards that provide comprehensive information about all bookings, schedules, resources, and customer requests. The frontend is developed using technologies such as **HTML, CSS, JavaScript, and modern frameworks like React or Angular**, ensuring responsive design for web and mobile platforms. This layer emphasizes usability and accessibility, allowing users to complete tasks efficiently.



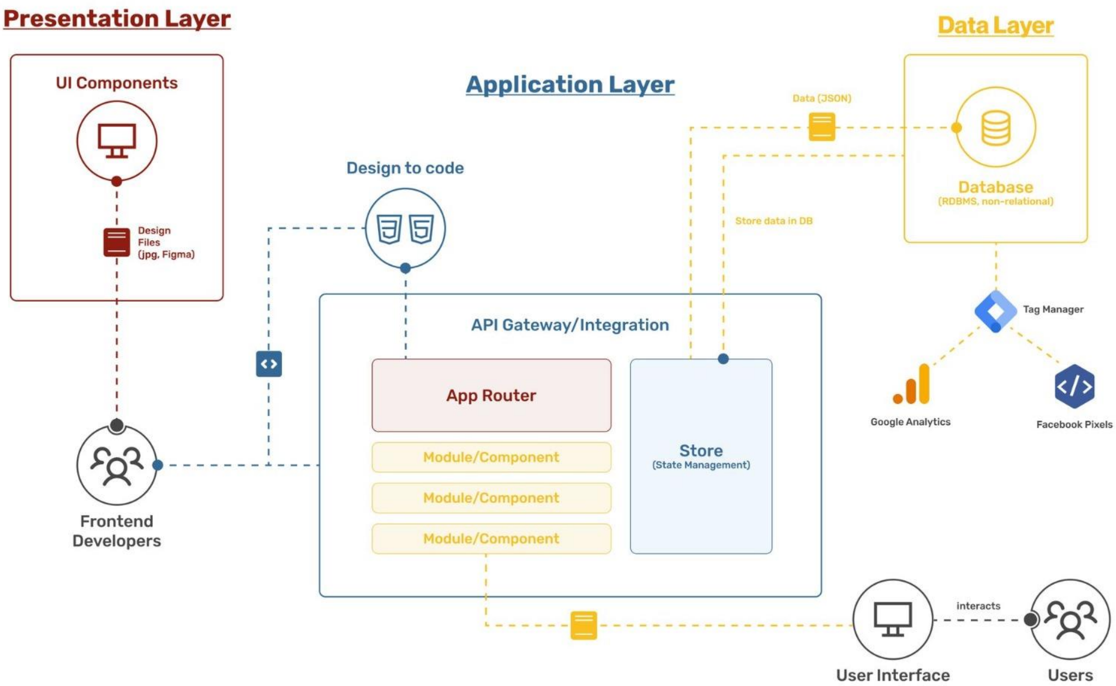
2. Application Layer (Backend)

The backend, or application layer, is the core of the system and handles **business logic, data processing, and communication between the frontend and database**. When a user initiates a booking, the backend validates the request, checks hall availability, calculates costs including additional services, and confirms the booking.

It also handles **payment processing** by integrating secure payment gateways, generates invoices, and ensures proper transaction records. The backend manages automated notifications for booking confirmations, reminders, and updates through email or SMS, keeping customers informed at every step.

For administrators, the backend provides tools for booking management, scheduling, staff allocation, pricing updates, and conflict resolution. Analytical modules process historical data to generate reports on revenue, booking trends, and customer preferences, supporting data-driven decision-making.

The backend is typically implemented using **server-side frameworks** such as **Node.js, Django, or Spring Boot**, which provide high performance, scalability, and integration capabilities with third-party services.

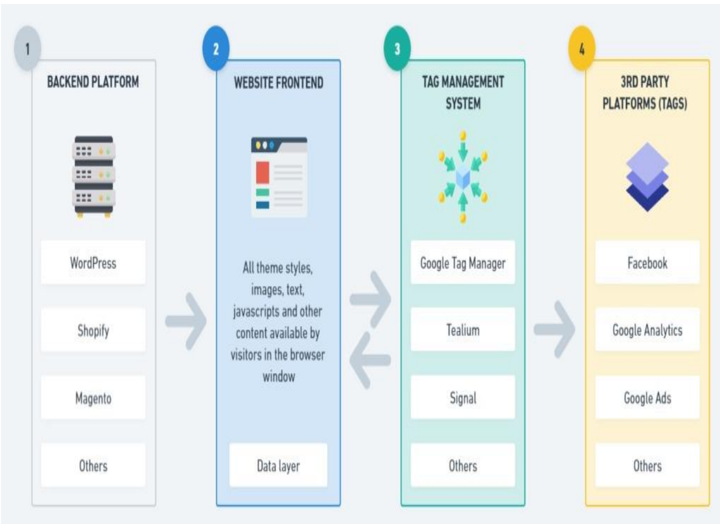


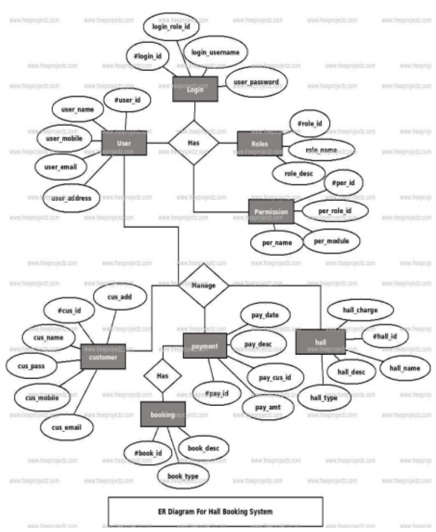
3. Data Layer (Database)

The database layer is responsible for storing and managing all essential information, including:

- Customer profiles and contact details
- Booking records and schedules
- Banquet hall details such as capacity, pricing, and facilities
- Payment and invoice records
- Staff and administrator data

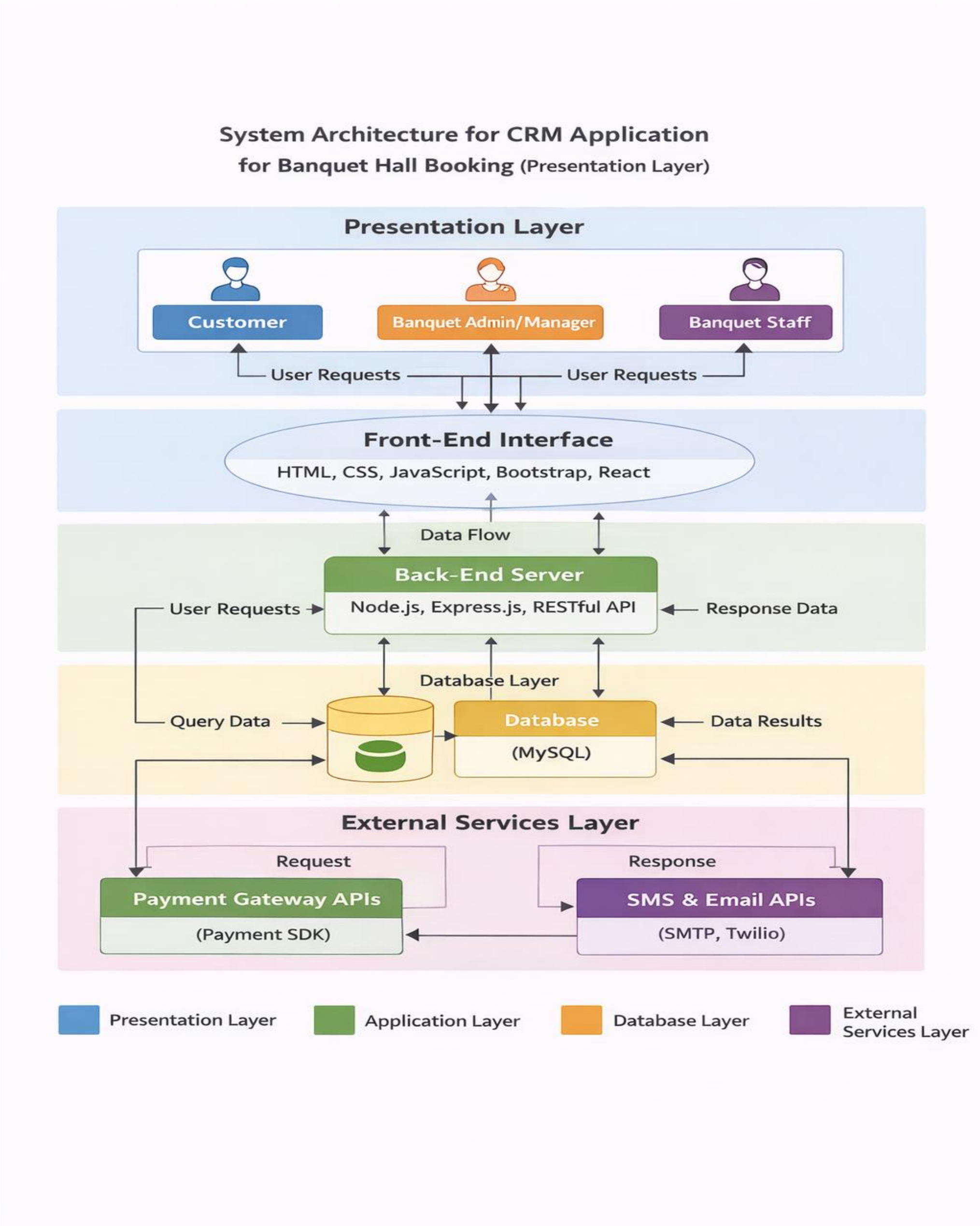
The database is designed using **relational database management systems (RDBMS)** like **MySQL or PostgreSQL**, which allow efficient data retrieval, secure storage, and easy reporting. Structured tables ensure that customer data, bookings, and payments are consistently organized, enabling personalized services and reliable analytics.





4. Additional Modules and Components

- **Authentication and Authorization:** Ensures secure access through role-based permissions for customers, administrators, and staff.
- **Notification Module:** Sends automated email and SMS alerts for booking confirmations, payment reminders, and event updates.
- **API Layer:** Facilitates integration with external services such as payment gateways, third-party vendors, and cloud platforms.
- **Reporting and Analytics Module:** Provides insights into business performance, including revenue trends, booking patterns, and customer preferences, supporting strategic planning.



5. Architecture Benefits

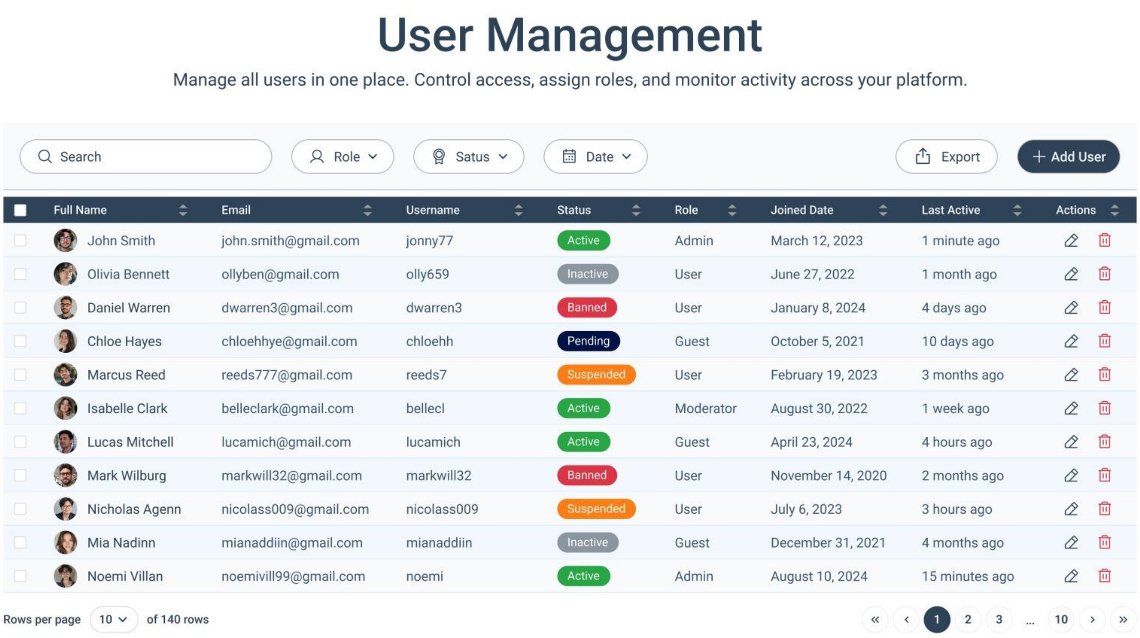
This architecture provides **real-time updates, seamless communication, high availability, and data security**. By separating concerns across layers, the system achieves **modularity, maintainability, and scalability**, allowing future enhancements such as mobile apps, AI-based recommendations, or vendor integrations.

In summary, the **CRM-based banquet hall booking system** relies on a well-structured architecture that ensures efficient user interactions, accurate data management, and smooth operational workflow. It integrates frontend usability, backend processing, secure data handling, and analytics to provide a **comprehensive and modern solution** for banquet hall management.

Key Modules

1. User Management

- **Registration & Login:** Users can create accounts with personal details, email verification, and password protection.
- **Role-Based Access Control:** Different users have distinct roles:
    - **Customers:** Can book halls, view booking history, and make payments.
    - **Administrators:** Can manage halls, approve bookings, assign staff, and view reports.
    - **Staff:** Can view assigned tasks, manage event preparations, and update progress.
- **Profile Management:** Users can update their personal details, contact information, and preferences.



Full Name	Email	Username	Status	Role	Joined Date	Last Active	Actions
John Smith	john.smith@gmail.com	jonny77	Active	Admin	March 12, 2023	1 minute ago	
Olivia Bennett	ollyben@gmail.com	olly659	Inactive	User	June 27, 2022	1 month ago	
Daniel Warren	dwarren3@gmail.com	dwarren3	Banned	User	January 8, 2024	4 days ago	
Chloe Hayes	chloehhye@gmail.com	chloehh	Pending	Guest	October 5, 2021	10 days ago	
Marcus Reed	reeds777@gmail.com	reeds7	Suspended	User	February 19, 2023	3 months ago	
Isabelle Clark	belleclark@gmail.com	bellecl	Active	Moderator	August 30, 2022	1 week ago	
Lucas Mitchell	lucamich@gmail.com	lucamich	Active	Guest	April 23, 2024	4 hours ago	
Mark Wilburg	markwill32@gmail.com	markwill32	Banned	User	November 14, 2020	2 months ago	
Nicholas Agenn	nicolass009@gmail.com	nicolass009	Suspended	User	July 6, 2023	3 hours ago	
Mia Nadinn	mianaddiin@gmail.com	mianaddiin	Inactive	Guest	December 31, 2021	4 months ago	
Noemi Villan	noemivill99@gmail.com	noemi	Active	Admin	August 10, 2024	15 minutes ago	

2. Customer Relationship Management (CRM)

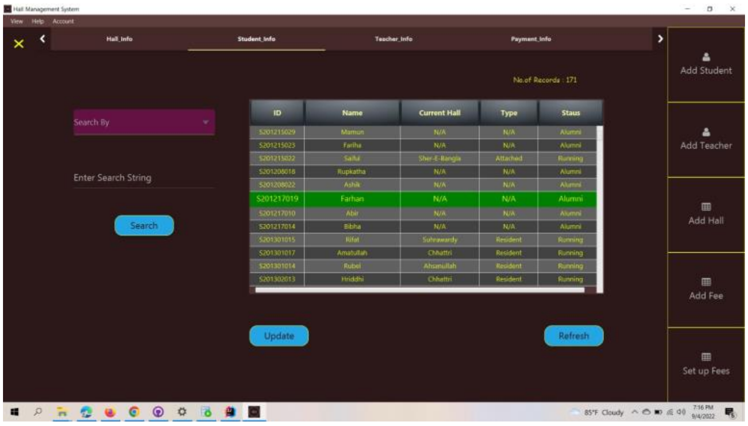
- **Customer Profiles:** Stores detailed customer information including contact details, past bookings, and preferences.
- **Booking History:** Keeps track of all previous bookings, including event type, date, and services used.
- **Feedback & Reviews:** Customers can provide feedback after events, helping improve services.
- **Personalized Recommendations:** System can suggest halls or services based on previous bookings and preferences.

3. Hall Management

- **Hall Details:** Maintains information on capacity, pricing, amenities, and photos.
- **Real-Time Availability:** Automatically updates hall availability to avoid conflicts.
- **Maintenance Scheduling:** Administrators can mark halls unavailable for maintenance or special purposes.
- **Pricing & Offers:** Supports dynamic pricing, seasonal discounts, and promotional offers.

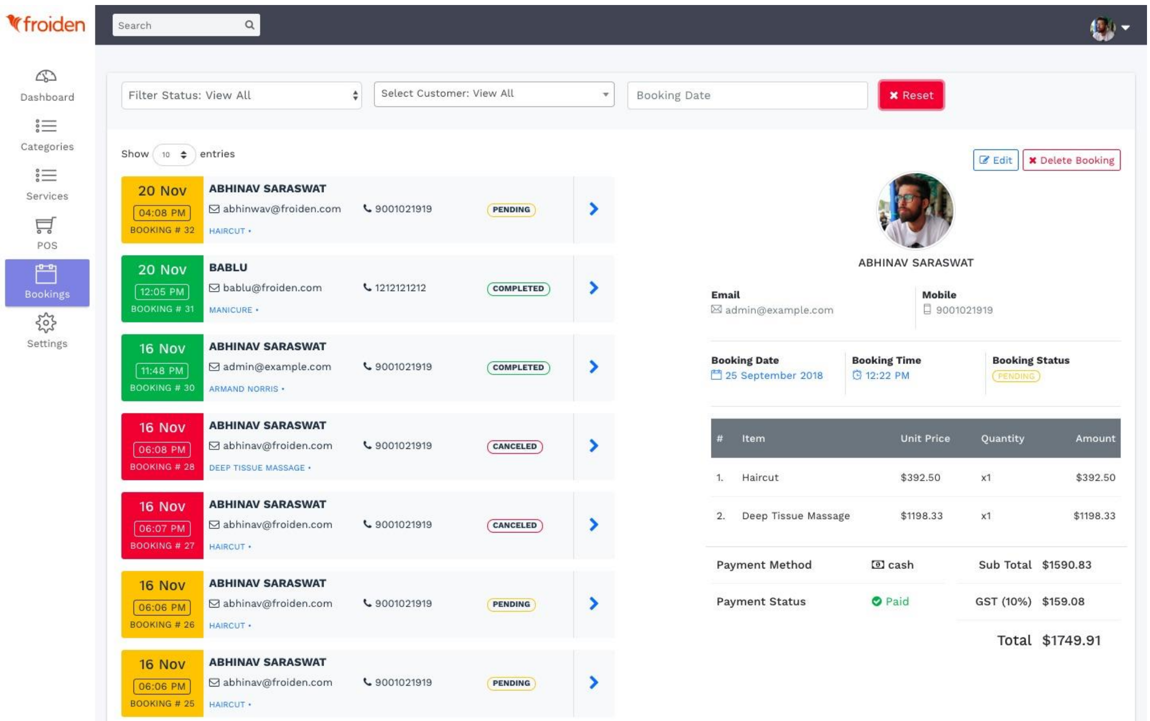
4. Booking Management

- **Reservation Creation:** Customers can select halls, dates, and services, then confirm bookings.
- **Modification & Cancellation:** Users can modify or cancel bookings, subject to policy rules.
- **Conflict Detection:** System prevents double bookings and schedules conflicts.
- **Staff Assignment:** Admins can assign staff to specific events for coordination.



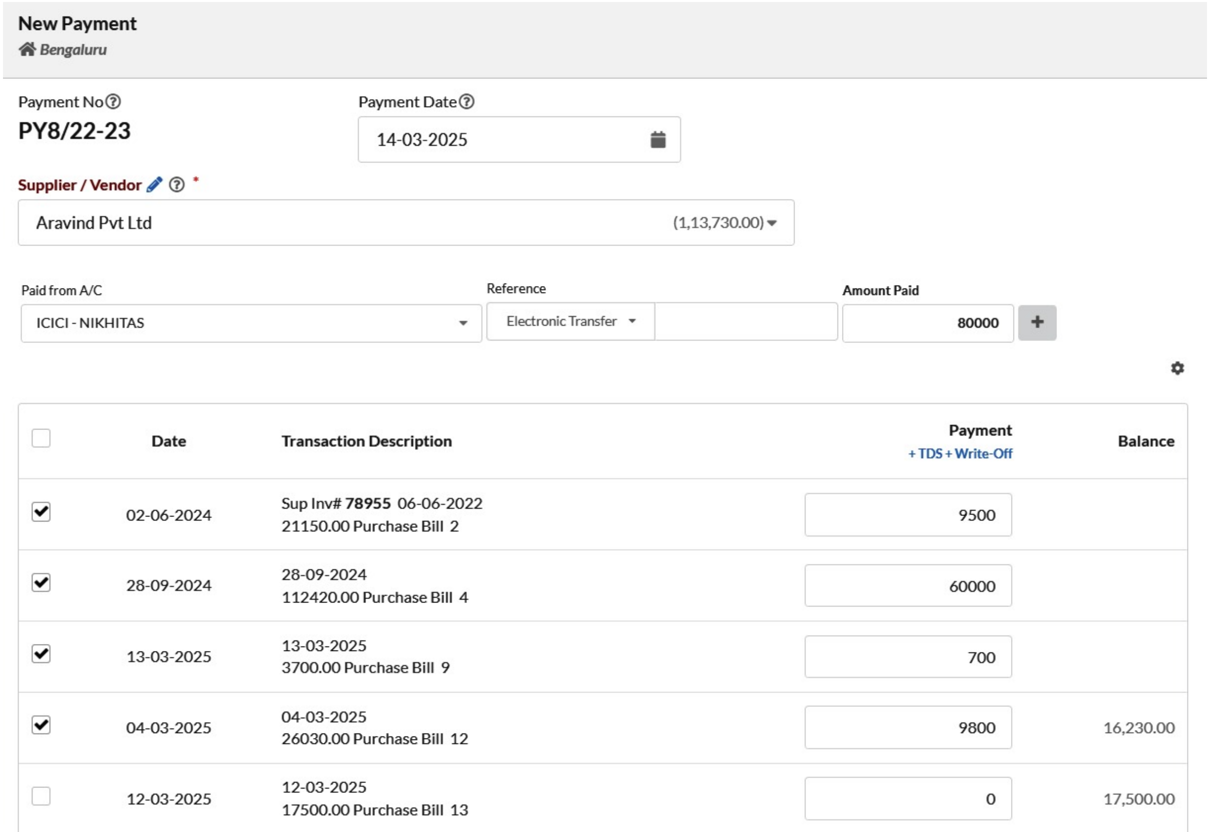
5. Event Management

- **Custom Event Planning:** Customers can specify event type (wedding, conference, party) and additional services like catering, decoration, and audio-visual setup.
- **Task Management for Staff:** Staff can access assigned tasks and mark progress, ensuring events are prepared according to customer specifications.
- **Resource Allocation:** System helps allocate chairs, tables, equipment, and other resources efficiently.



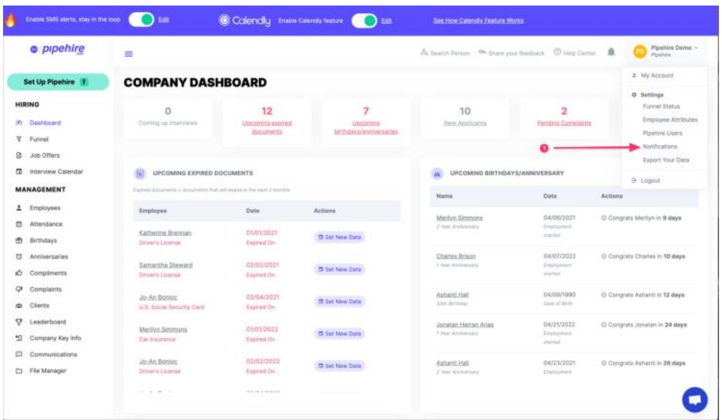
6. Payment and Billing

- **Secure Payment Integration:** Supports online payments via gateways like Stripe or Razorpay.
- **Invoice Generation:** Automatically generates invoices for bookings and services.
- **Payment Tracking:** Keeps record of pending and completed payments.
- **Refund Handling:** Supports refund processing for cancelled bookings.



7. Notification Module

- **Booking Confirmations:** Sends instant confirmation via email or SMS after booking.
- **Reminders & Alerts:** Automated notifications for payment due dates, event reminders, or schedule changes.
- **Status Updates:** Staff and admins receive alerts on booking approvals, task assignments, and event updates.



8. Reporting and Analytics

- **Revenue Reports:** Generates daily, weekly, or monthly revenue summaries.
- **Booking Trends:** Tracks popular dates, hall utilization, and seasonal demand.
- **Customer Insights:** Analyses customer behaviour and preferences for targeted promotions.
- **Operational Reports:** Helps admins monitor staff performance, task completion, and overall efficiency.

9. Integration & API Features

- **Payment Gateways:** Securely processes online transactions.
- **Communication Services:** Connects with email and SMS providers for automated notifications.
- **Cloud Storage Integration:** Optional integration for storing media files, invoices, or event photos.

10. Summary

The functional requirements ensure that the CRM system is **user-friendly, efficient, and comprehensive**. Customers can book halls and manage events easily, administrators can oversee operations and generate reports, and staff can efficiently execute event preparations. This modular approach guarantees seamless communication, accurate data management, and improved business performance.

Functional Requirements

Functional requirements define **what the system must do**, describing the services, tasks, and operations it must perform to meet the needs of customers, administrators, and staff. The CRM-based banquet hall booking system is designed to streamline booking processes, manage customer relationships, and enhance operational efficiency.

1. User Management

- **Registration and Login:** Users must be able to create accounts, verify their email, and securely log in.
- **Role-Based Access Control:** The system distinguishes between three types of users:
    - **Customers:** Can browse halls, make bookings, view booking history, and make payments.
    - **Administrators:** Can manage halls, approve bookings, assign staff, and view reports.
    - **Staff:** Can view assigned tasks and manage event preparations.
- **Profile Management:** Users can update personal details, contact information, and preferences.

2. Customer Relationship Management (CRM)

- **Customer Profiles:** Stores contact details, booking history, and preferences.
- **Booking History Tracking:** Maintains a record of all past and ongoing bookings.
- **Feedback Management:** Allows customers to provide reviews after events.
- **Personalized Recommendations:** Suggests halls or services based on customer preferences and past bookings.

3. Hall Management

- **Hall Information:** Stores details such as capacity, pricing, amenities, and images.
- **Real-Time Availability:** Updates Hall availability to prevent double bookings.
- **Maintenance Scheduling:** Administrators can block halls for maintenance or special use.
- **Pricing Management:** Supports dynamic pricing, discounts, and promotional offers.

4. Booking Management

- **Reservation Creation:** Customers can select halls, dates, and additional services to create a booking.
- **Modification and Cancellation:** Allows bookings to be updated or cancelled according to policy rules.
- **Conflict Detection:** Prevents double bookings and scheduling conflicts.
- **Staff Assignment:** Admins can assign staff to specific events.

5. Event Management

- **Custom Event Planning:** Customers can select services like catering, decoration, and seating arrangements.
- **Task Management:** Staff can view and update the status of assigned tasks.
- **Resource Allocation:** Helps allocate furniture, equipment, and other resources efficiently.

6. Payment and Billing

- **Secure Payment Integration:** Supports online payments via secure gateways.
- **Invoice Generation:** Automatically generates invoices for bookings and services.
- **Payment Tracking:** Tracks pending, completed, and refunded payments.
- **Refund Processing:** Handles refunds for cancelled bookings.

7. Notification Module

- **Booking Confirmations:** Sends instant confirmations via email or SMS.
- **Reminders and Alerts:** Notifies users about payment dues, upcoming events, or schedule changes.
- **Status Updates:** Informs staff and admins of booking approvals and task assignments.

8. Reporting and Analytics

- **Revenue Reports:** Tracks daily, weekly, and monthly revenue.
- **Booking Trends:** Identifies peak seasons and hall utilization.
- **Customer Insights:** Analyses preferences and behaviour for marketing strategies.
- **Operational Reports:** Monitors staff performance, event readiness, and task completion.

9. Summary

The functional requirements ensure that the system is **efficient, user-friendly, and reliable**. Customers can book halls and manage events seamlessly, administrators can oversee operations and make data-driven decisions, and staff can execute tasks efficiently. This structure provides **streamlined operations, accurate data management, and improved customer satisfaction**, forming the backbone of the CRM-based banquet hall booking system.

<u>Non-Functional Requirements</u>

Non-functional requirements define **how the system should perform**, specifying qualities, constraints, and standards rather than specific functions. They ensure the CRM application operates reliably, efficiently, securely, and provides a positive user experience.

1. Performance Requirements

- The system should handle **at least 500 simultaneous users** without performance degradation.
- Booking operations, search queries, and payment processing should complete within **3–5 seconds**.
- Real-time updates for hall availability should reflect immediately to prevent double bookings.

Performance measures
Repair (R): Expected cost for component/content repair/replacement, μ_R (percentage of building value including contents)
Downtime (D): Expected closure of the building for repair/replacement, μ_D (days)
Injury (I): Expected likelihood of injury/casualty, μ_I (probability of minor/major injury and death)

		Allowable loss (Capacity)		
Ground motion intensity corresponding to	Expected loss (Demand)	Residential buildings	Commercial and office buildings	Emergency facilities
Frequently occurring earthquake (FOE), 50% in 50 yrs	Repair: μ_R	1%	0.1%	0.1%
	Downtime: μ_D	1 day	0.1 day	0.1 day
	Injury: μ_I	1%,0.1%,0.1%	0.1%,0.1%,0.1%	0.1%,0.1%,0.1%
Design basis earthquakes (DBE), 10% in 50 yrs	Repair: μ_R	10%	5%	1%
	Downtime: μ_D	5 days	1 day	0.1 day
	Injury: μ_I	10%,2%,0.1%	5%,1%,0.1%	1%,0.1%,0.1%
Maximum considered earthquake (MCE), 2% in 50 yrs	Repair: μ_R	No limit	10%	5%
	Downtime: μ_D	No limit	5 days	1 day
	Injury: μ_I	50%,10%,1%	10%,2%,0.1%	5%,1%,0.1%

2. Scalability

- The system must be scalable to accommodate **an increasing number of users, halls, and bookings**.
- Cloud-based deployment or modular architecture should allow easy addition of new features, halls, and services.

3. Reliability & Availability

- The system should ensure **99% uptime**, especially during peak booking periods.
- Automatic backup mechanisms must store booking, payment, and user data to prevent data loss.
- Failover mechanisms should maintain operations during system failures.

Aspect	Availability	Reliability
Definition	The ability of a system to be operational and accessible when needed.	The ability of a system to perform its intended function consistently over time without failure.
Focus	Short-term: Ensuring systems are up and running at any given time.	Long-term: Ensuring consistent and dependable system performance over time.
Key Metric	Uptime percentage (e.g., "99.99% uptime").	MTBF (Mean Time Between Failures), failure rates.
Impact of Downtime	Immediate access is disrupted (e.g., customers can't access services).	System failures impact overall dependability and functionality.
Examples	Cloud platforms, e-commerce websites, streaming services.	Industrial machinery and safety-critical devices like airplane systems.
Primary Goal	Maximize operational time (reducing downtime).	Minimize system failures (increasing dependability).
Trade-offs	Redundancy improves availability but may not enhance reliability.	Overengineering reliability can reduce availability during maintenance.
Critical Industries	Customer-facing services (e.g., banking, healthcare access).	Manufacturing, aerospace, and medical devices.
Challenges	Maintaining uptime amidst unexpected issues like power outages.	Preventing frequent breakdowns or malfunctions over time.

4. Security Requirements

- Secure authentication with **role-based access control** for customers, administrators, and staff.
- All sensitive data (passwords, payment details) must be **encrypted using SSL/TLS and AES**.
- The system should comply with **data protection regulations** to ensure customer privacy.
- Protection against SQL injection, XSS attacks, and unauthorized access.

5. Usability

- The system should have an **intuitive and responsive user interface** for both web and mobile platforms.
- Customers should be able to complete bookings with **minimal steps**.
- Administrators and staff should have dashboards that allow easy monitoring of bookings and events.

6. Maintainability

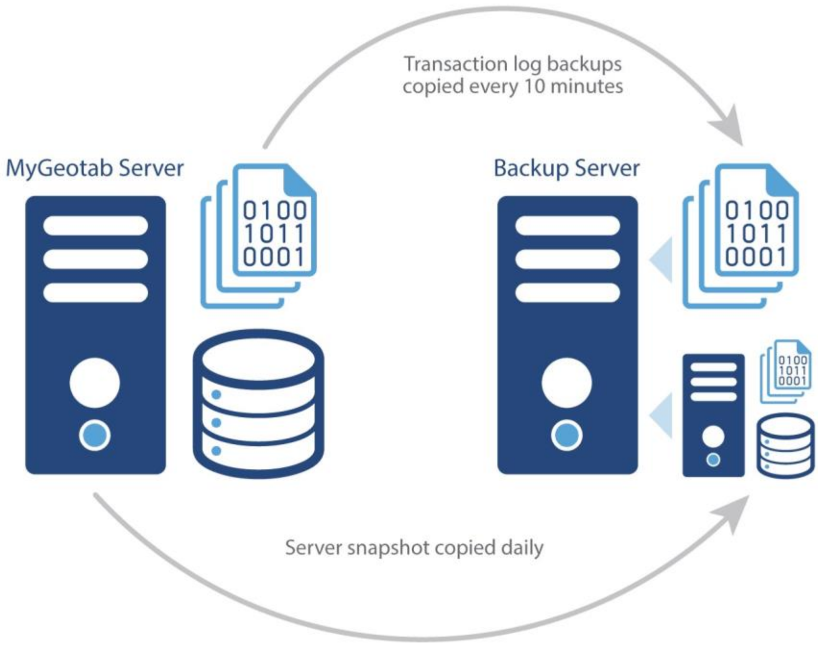
- The system code should be modular, following **best practices for readability and reuse**.
- Updates, bug fixes, or feature additions should be implementable without affecting existing functionality.
- Comprehensive documentation must be available for future developers.

7. Compatibility

- The system should be compatible with **major web browsers** (Chrome, Firefox, Edge, Safari).
- The mobile interface should support **both Android and iOS devices**.

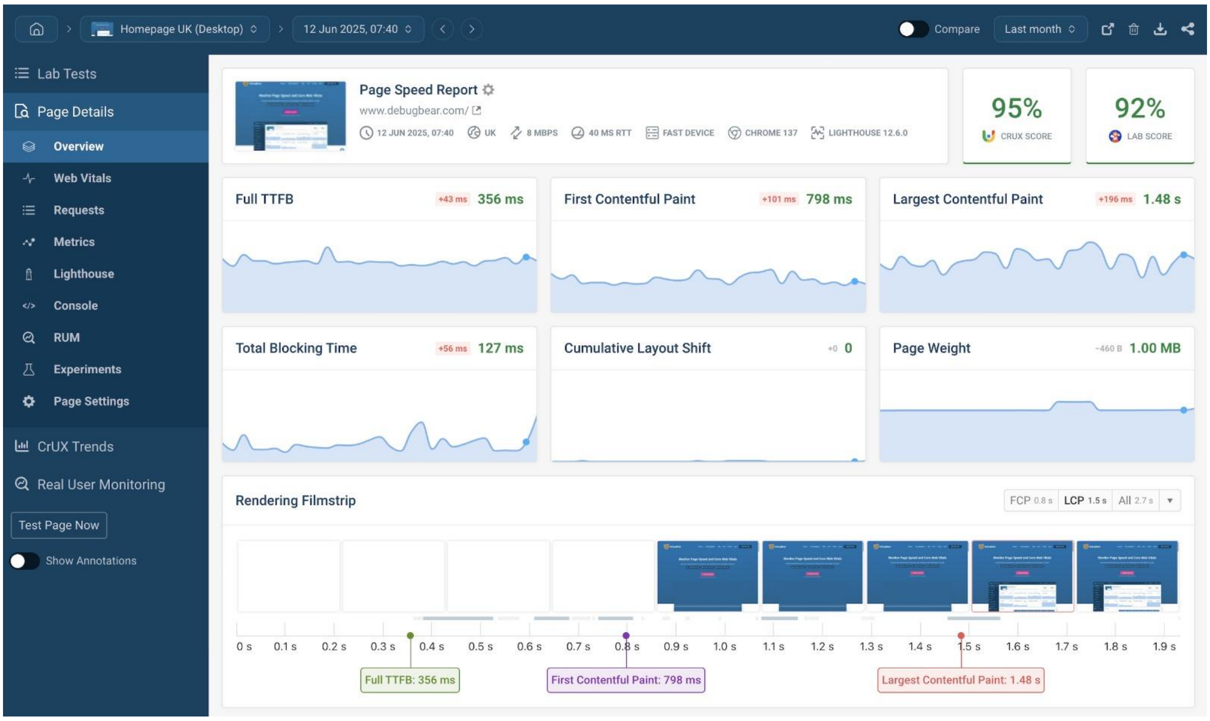
8. Backup & Recovery

- Automatic daily backups of database and system logs.
- The system should provide a **disaster recovery mechanism** to restore data within a short period.



9. Performance Monitoring & Analytics

- System must include monitoring tools to track **server performance, user activity, and booking trends**.
- Administrators should receive alerts if the system experiences performance bottlenecks.



10. Summary

Non-functional requirements ensure that the CRM application is **secure, fast, reliable, and user-friendly**. While functional requirements define what the system does, these NFRs guarantee **quality, efficiency, and user satisfaction**, making the system suitable for real-world banquet hall management operations.

Database Design

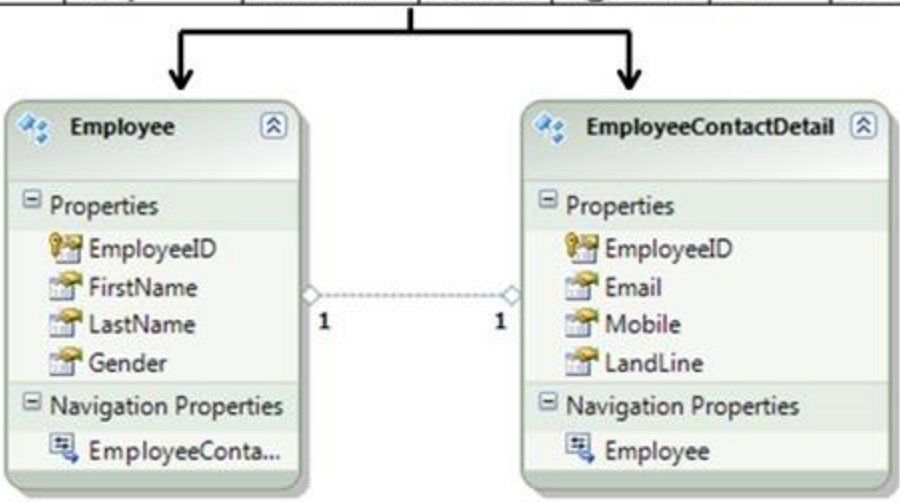
The **Database Design** is a fundamental aspect of the CRM Application for Banquet Hall Booking. It defines how all data—related to customers, banquet halls, bookings, payments, staff, and events—is stored, managed, and retrieved efficiently. A well-designed database ensures **data integrity, security, scalability, and performance**, which are crucial for a system that manages multiple users, real-time bookings, and financial transactions.

The system uses a **Relational Database Management System (RDBMS)**, such as **MySQL or PostgreSQL**, to maintain structured data with relationships that reflect real-world operations. The design follows **normalization principles** to eliminate redundancy, ensure consistency, and improve query efficiency.

1. Entities and Tables

The database contains multiple entities, each represented as a table, which corresponds to a specific aspect of the system. The key tables are:

EmployeeID	FirstName	LastName	Gender	Email	Mobile	LandLine
1	Mark	Hastings	Male	x@x.com	XXX	XXX
2	Steve	Pound	Male	y@y.com	YYY	YYY
3	Ben	Hoskins	Male	z@z.com	ZZZ	ZZZ
4	Philip	Hastings	Male	a@a.com	AAA	AAA
5	Mary	Lambeth	Female	b@b.com	BBB	BBB



1.1 User Table

The **User Table** stores general information about all system users, including customers, administrators, and staff. It provides authentication and role management, forming the foundation for security and access control.

Attributes:

- `user_id` (Primary Key) – Unique identifier for each user.
- `name` – Full name of the user.
- `email` – Used for login and notifications.
- `password` – Encrypted password for authentication.
- `role` – Defines whether the user is a Customer, Admin, or Staff.
- `phone_number` – Contact number for communication.
- `created_at` – Account creation timestamp.
- `updated_at` – Last updated timestamp.

Rationale: This table centralizes user credentials and roles, allowing the system to enforce **role-based access control** effectively.

1.2 Customer Table

While all customers are users, the **Customer Table** stores additional CRM-related details, enabling personalized service and tracking booking history.

Attributes:

- `customer_id` (Primary Key)
- `user_id` (Foreign Key → User.user_id)
- `address` – Residential or billing address.
- `preferences` – Event types or services favored by the customer.
- `loyalty_points` – Rewards points for loyalty programs.
- `created_at`
- `updated_at`

Rationale: By separating CRM-specific details, the system can analyse customer behaviour, offer tailored promotions, and improve engagement without cluttering the main user table.

1.3 Banquet Hall Table

This table stores all relevant information about the banquet halls managed by the system.

Attributes:

- `hall_id` (Primary Key)
- `hall_name` – Name of the banquet hall.
- `location` – Physical address.
- `capacity` – Maximum number of guests.
- `pricing` – Base rental cost.
- `amenities` – Facilities provided (e.g., AV equipment, catering).
- `availability_status` – Available/Occupied/Maintenance.
- `created_at`
- `updated_at`

Rationale: The hall table enables customers to view availability, compare options, and select halls based on capacity and amenities. Real-time availability tracking is critical to prevent double bookings.

1.4 Booking Table

The **Booking Table** is the core of the system, connecting customers, halls, and event details.

Attributes:

- `booking_id` (Primary Key)
- `customer_id` (Foreign Key → Customer.customer_id)
- `hall_id` (Foreign Key → BanquetHall.hall_id)
- `event_date` – Date and time of the event.
- `event_type` – Wedding, conference, party, etc.
- `services_selected` – JSON or comma-separated string of additional services like catering or decoration.
- `booking_status` – Confirmed, Pending, or Cancelled.
- `created_at`
- `updated_at`

Rationale: This table provides a **single source of truth for all bookings**, linking customers to halls while tracking the status and event-specific details.

1.5 Payment Table

This table manages all financial transactions associated with bookings.

Attributes:

- `payment_id` (Primary Key)
- `booking_id` (Foreign Key → Booking.booking_id)
- `amount` – Total payment amount.
- `payment_method` – Credit Card, Debit Card, UPI, Wallet.
- `payment_status` – Paid, Pending, or Refunded.
- `transaction_date` – Timestamp of payment completion.

Rationale: Payment tracking ensures financial transparency and allows automated invoice generation and refunds. Linking payments to bookings provides traceability.

1.6 Staff Table

Staff members assigned to events are managed in this table.

Attributes:

- `staff_id` (Primary Key)
- `user_id` (Foreign Key → User.user_id)
- `assigned_tasks` – Tasks for each event.
- `availability_status` – Available, Busy, or Off-duty.
- `created_at`
- `updated_at`

Rationale: This enables efficient resource allocation, ensuring staff is properly assigned and events are prepared on time.

1.7 Feedback Table

Customer reviews and ratings are stored here to improve service quality.

Attributes:

- `feedback_id` (Primary Key)
- `customer_id` (Foreign Key → Customer.customer_id)
- `booking_id` (Foreign Key → Booking.booking_id)
- `rating` – Numeric score (1–5).
- `comments` – Textual feedback.
- `created_at`

Rationale: Feedback analysis helps management identify strengths, weaknesses, and customer preferences.

2. Relationships Between Tables

- **User → Customer / Staff:** One-to-one relationship. A single user can be either a customer or staff.
- **Customer → Booking:** One-to-many relationship. A customer can make multiple bookings over time.
- **Banquet Hall → Booking:** One-to-many relationship. Each hall can host multiple events, but only one per timeslot.
- **Booking → Payment:** One-to-one or one-to-many relationship depending on instalment options.
- **Customer → Feedback:** One-to-many relationship. A customer can provide feedback for multiple bookings.

3. Normalization

The database follows **3NF (Third Normal Form)**:

- **1NF:** All tables have atomic values.
- **2NF:** All non-key attributes fully depend on the primary key.
- **3NF:** No transitive dependency exists.

This eliminates redundancy, improves consistency, and ensures **efficient data retrieval** for queries like hall availability, booking history, and revenue reports.

4. Indexes and Keys

- Primary keys ensure uniqueness for each table.
- Foreign keys maintain **referential integrity** between related tables.
- Indexes on columns like event_date, hall_id, and customer_id speed up queries for availability checks and reports.

5. Security Considerations

- **Encryption:** Passwords and sensitive data are encrypted.
- **Access Control:** Role-based access ensures only authorized users can modify sensitive data.
- **Backup:** Regular automated backups prevent data loss.

6. Rationale and Benefits

A relational database design provides:

- **Data Integrity:** Prevents duplication and maintains consistency.
- **Scalability:** Easy to expand with new halls, customers, or services.
- **Efficiency:** Supports real-time queries for bookings and payments.
- **Analytics:** Facilitates reporting, revenue tracking, and customer behaviour analysis.

7. Summary

The database design forms the backbone of the CRM Application for Banquet Hall Booking. By maintaining **well-structured tables, normalized relationships, and secure storage**, it ensures efficient booking management, accurate payment tracking, effective staff allocation, and personalized customer service. This design also supports scalability, real-time performance, and data-driven decision-making, making it suitable for modern banquet hall operations.

Workflow

The CRM application for banquet hall booking follows a structured workflow that integrates both **data relationships (ER Diagram)** and **process flow (DFD)** to ensure efficient management of customer interactions, bookings, and event execution. This workflow begins with customer inquiry and continues through booking confirmation, event planning, billing, and feedback collection.

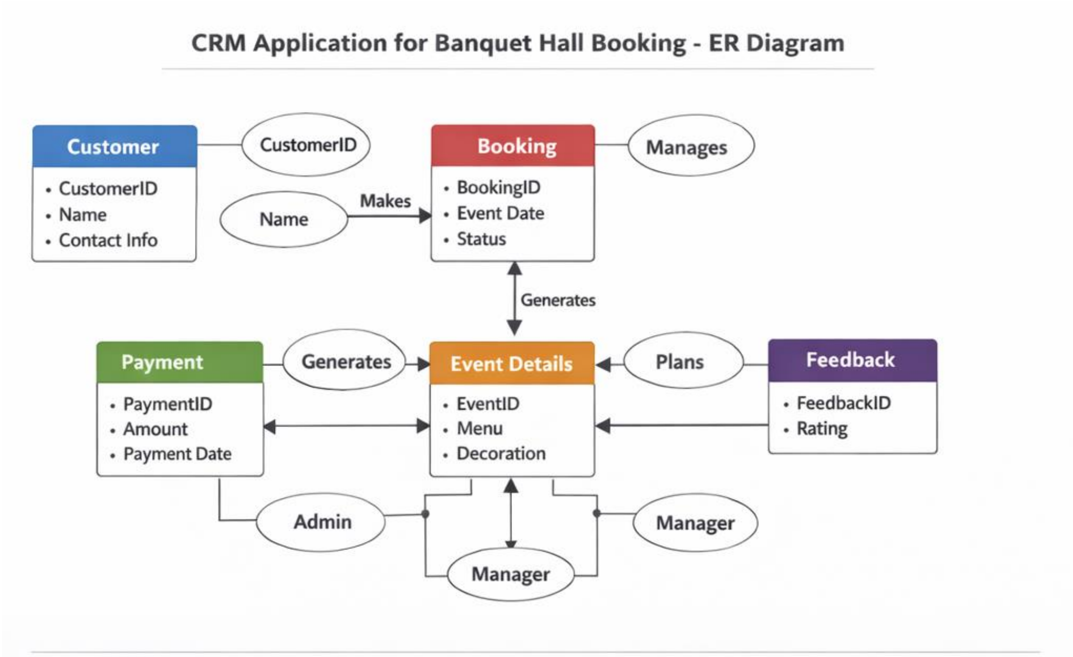
The process starts with the **customer inquiry stage**, where a customer expresses interest in booking a banquet hall. The inquiry may come through various channels such as a website form, phone call, or direct visit. The system collects essential details including customer name, contact information, event type, preferred date, and number of guests. These details are stored in the CRM system and treated as initial input data.

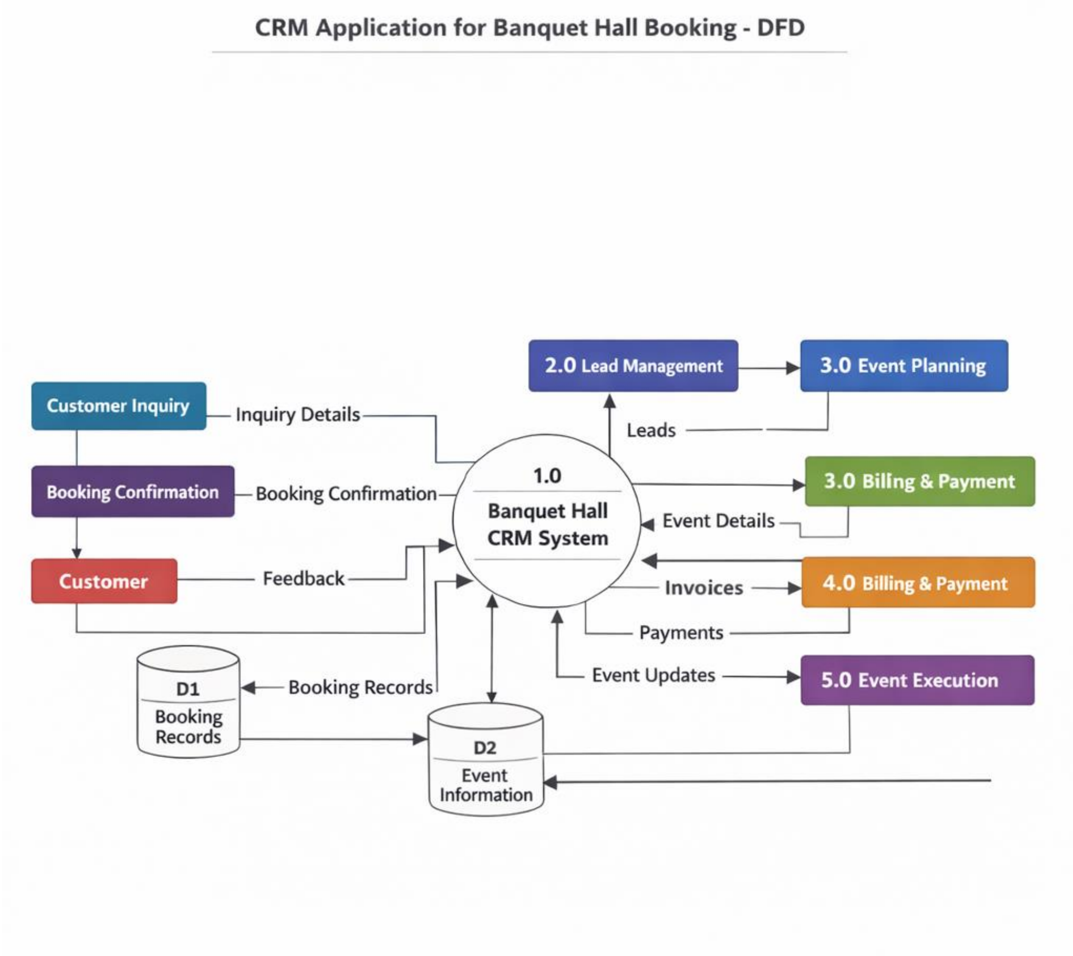
Next, the system moves to the **lead management stage**, as shown in the DFD. Here, the inquiry is converted into a lead and assigned to a booking manager. The manager reviews the request, checks hall availability, and communicates with the customer to understand requirements in detail. Based on the interaction, the lead is categorized into statuses such as new, interested, or rejected. If the customer shows interest, the process proceeds to booking confirmation.

During the **booking confirmation stage**, the system verifies the availability of the banquet hall for the requested date and time. Once confirmed, a quotation is generated including hall charges, catering, and additional services. After customer approval, an advance payment is made, and the booking is officially confirmed. At this point, the booking data is stored in the **Booking Records (D1)** data store. In the ER diagram, this corresponds to the relationship where the **Customer entity creates a Booking entity**, establishing a strong link between customer and reservation.

Following confirmation, the workflow enters the **event planning stage**. In this phase, detailed event requirements are collected, such as menu selection, decoration preferences, seating arrangements, and technical setups. These details are stored in the **Event Information (D2)** data store. According to the ER diagram, the **Booking entity generates Event Details**, which are then managed by system users like admins or managers. This ensures that all event-related data is centralized and easily accessible.

The next stage is **billing and payment processing**, where the system generates invoices based on selected services and event requirements. Customers are provided with multiple payment options such as cash, card, or online transactions. Payments are recorded and linked to the respective booking. In the ER diagram, this is represented by the **Payment entity**, which is associated with both booking and event details, ensuring accurate financial tracking.





Once payments are processed, the workflow proceeds to the **event execution stage**. The operations team uses the stored event information to carry out the event as planned. This includes arranging catering, decorations, and managing the event schedule. The CRM system supports this stage by providing real-time access to event details and allowing updates if any changes occur during execution.

After the event is completed, the final stage is **feedback collection and follow-up**. The system requests feedback from the customer regarding their experience. This feedback is stored in the **Feedback entity**, which is linked to the customer and event in the ER diagram. The collected feedback helps improve future services and maintain customer relationships.

In conclusion, the workflow of the CRM application ensures smooth coordination between different stages of banquet hall booking. The **DFD illustrates how data flows through various processes**, while the **ER diagram defines how data entities are interconnected**. Together, they provide a comprehensive system that improves efficiency, reduces manual errors, and enhances customer satisfaction.

Booking Process Flow:

1. Customer logs in
2. Selects hall and date
3. System checks availability
4. Customer confirms booking
5. Payment is processed
6. Booking confirmation sent

Technology Stack (Example)

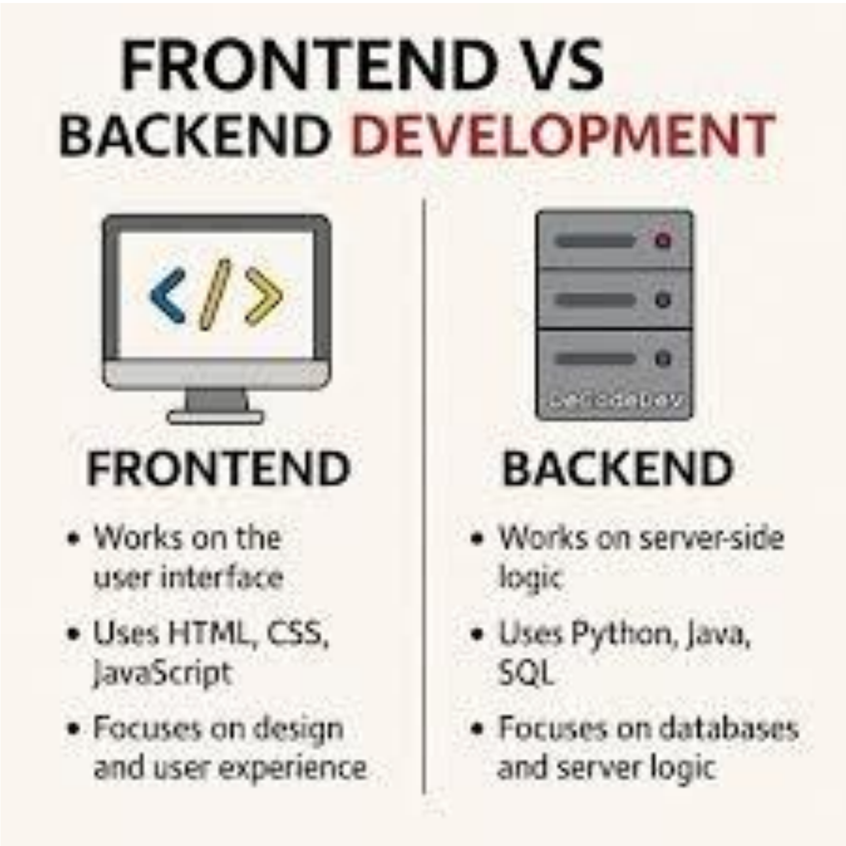
1. Frontend (User Interface)

The frontend is responsible for the user interface and user experience.

- **HTML5** – Structure of web pages
- **CSS3** – Styling and layout design
- **JavaScript** – Client-side interactivity
- **Bootstrap** – Responsive design for mobile and desktop
- *(Optional Advanced)*:
    - **React.js** – For dynamic and fast UI development

Features:

- User-friendly booking forms
- Dashboard for admin and managers
- Responsive design for all devices



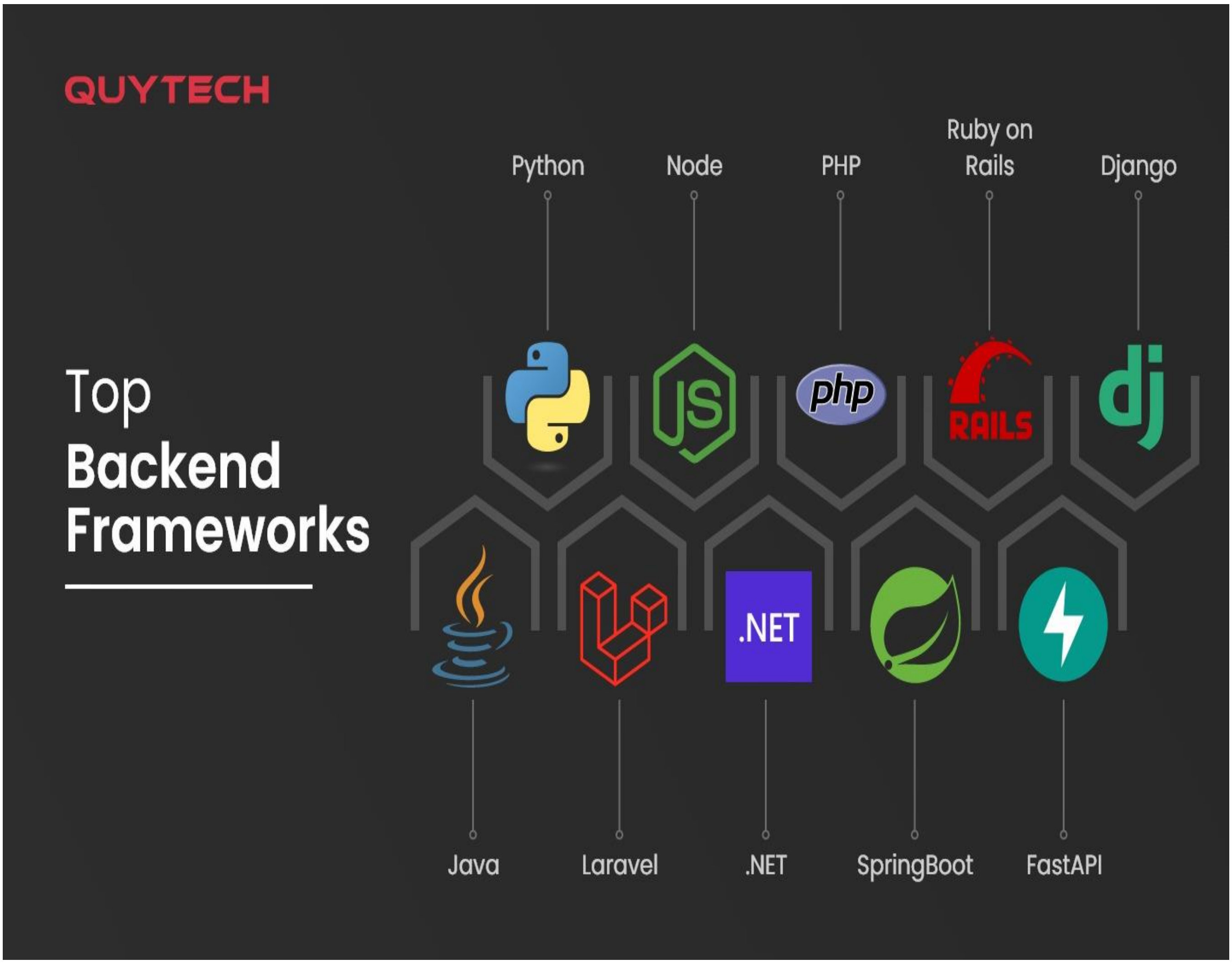
2. Backend (Server-Side Logic)

The backend handles business logic, authentication, and data processing.

- **Node.js** with **Express.js** *(or)*
- **Python (Django / Flask)** *(or)*
- **Java (Spring Boot)**

Responsibilities:

- Handling customer requests
- Processing bookings
- Managing authentication and authorization
- API development



3. Database (Data Storage)

The database stores all system data such as customers, bookings, payments, and events.

- **MySQL** – Relational database (most common for projects)
- *(Alternative)*:
    - **PostgreSQL**
    - **MongoDB** (NoSQL option)

Stored Data:

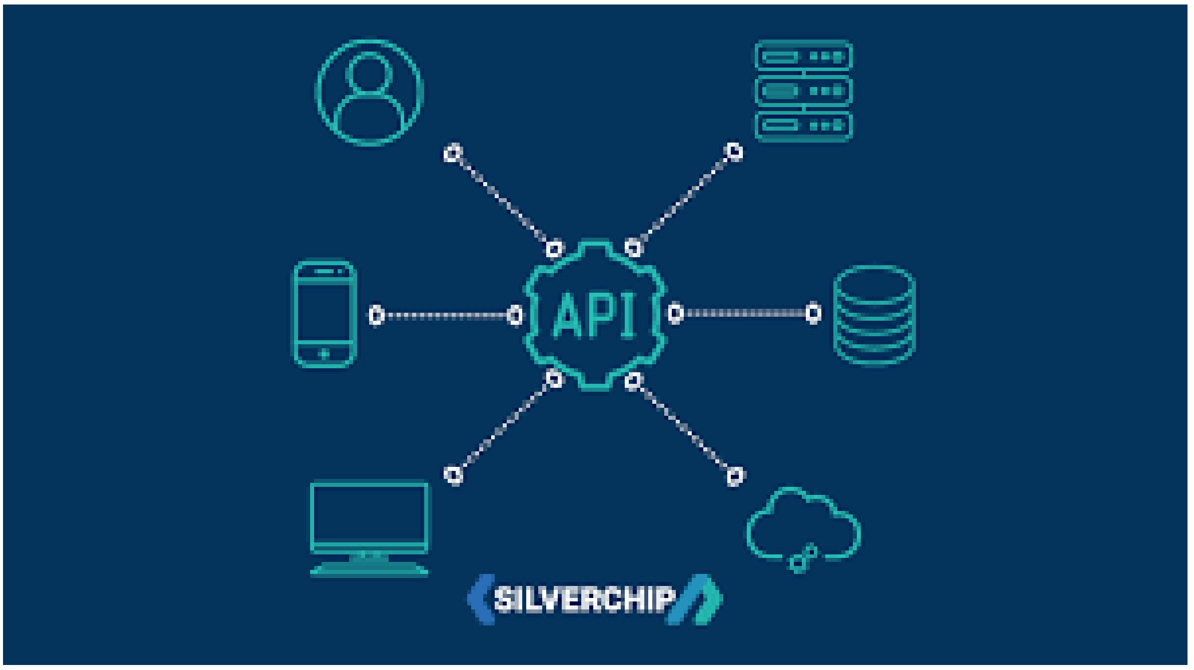
- Customer details
- Booking records
- Event information
- Payment history
- Feedback

4. CRM Functional Modules

- Lead Management Module
- Booking Management Module
- Event Planning Module
- Billing & Payment Module
- Feedback Management Module

5. APIs & Integrations

- **Payment Gateway APIs** (Razorpay / Stripe)
- **Email Services** (SMTP / Gmail API)
- **SMS Notifications API**

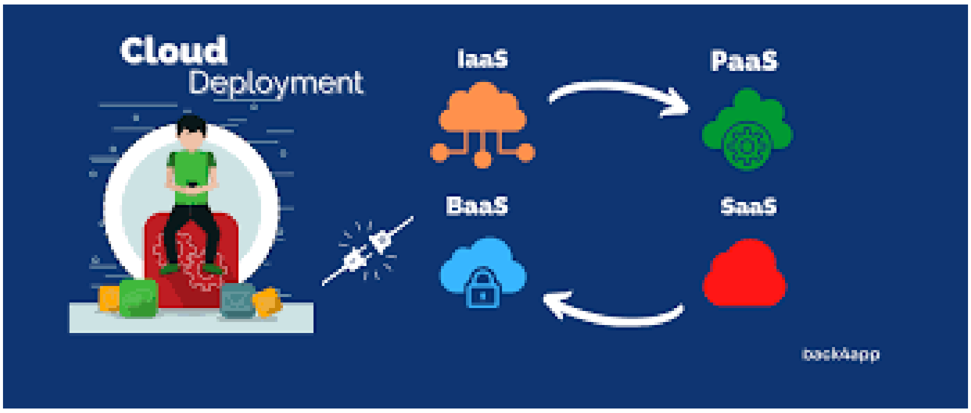


6. Development Tools

- **Visual Studio Code** – Code editor
- **Git & GitHub** – Version control
- **Postman** – API testing
- **XAMPP / WAMP** – Local server (for PHP-based systems)

7. Deployment Platform

- **Frontend Hosting**: Netlify / Vercel
- **Backend Hosting**: Heroku / Render
- **Database Hosting**: AWS RDS / Firebase



8. Security Technologies

- **JWT (JSON Web Tokens)** – Secure authentication
- **HTTPS** – Secure communication
- **Role-Based Access Control (RBAC)**

9. Optional Advanced Technologies

- **Cloud Storage (AWS S3)** – Store images/documents
- **Docker** – Containerization
- **AI Chatbot** – Customer support automation

10. Summary

- **Frontend:** HTML, CSS, JavaScript, React
- **Backend:** Node.js / Django / Spring Boot
- **Database:** MySQL / PostgreSQL
- **Cloud:** AWS / Azure
- **Payment Integration:** Razorpay / Stripe

Security Features

Security is a critical aspect of the CRM application, as it handles sensitive customer data, booking details, and financial transactions. The system implements multiple layers of security to ensure data protection, privacy, and safe operations.

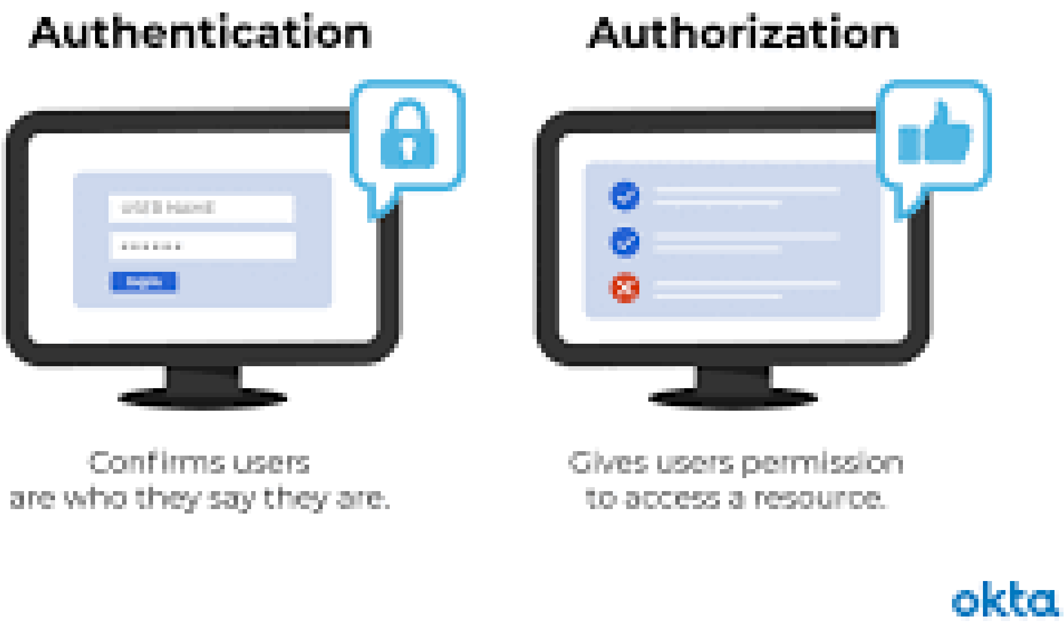
1. User Authentication

The system uses secure authentication mechanisms to verify user identity.

- Users must log in using a **username and password**
- Passwords are stored in **encrypted (hashed) format**
- Multi-factor authentication (optional) can be implemented using OTP (One-Time Password)

Benefit:

Prevents unauthorized users from accessing the system.



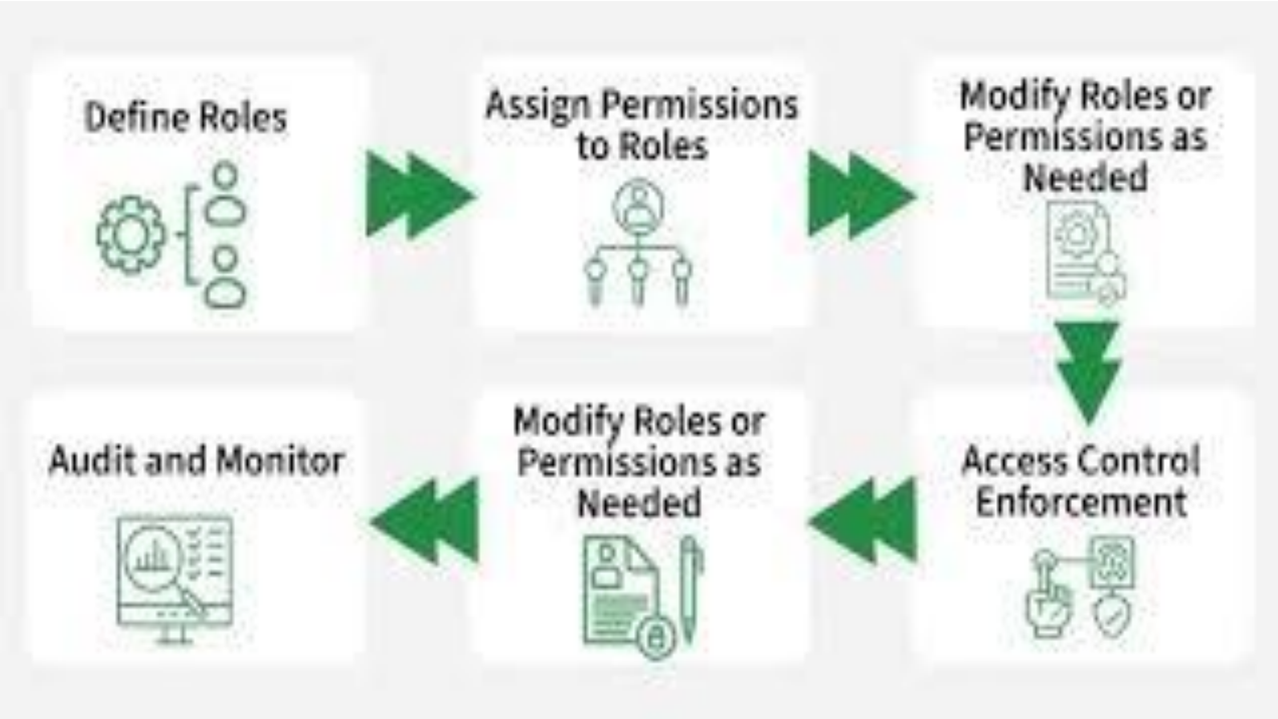
2. Role-Based Access Control (RBAC)

Access to system features is restricted based on user roles.

- **Admin** – Full access to all modules
- **Manager** – Access to bookings, leads, and event planning
- **Staff** – Limited access to assigned tasks
- **Customer** – Access only to their bookings

Benefit:

Ensures that users can only access relevant data, reducing misuse.



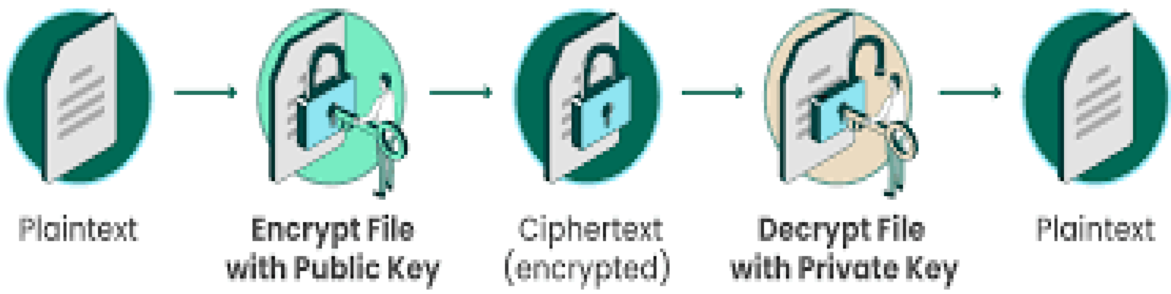
3. Data Encryption

Sensitive data is protected using encryption techniques.

- **HTTPS (SSL/TLS)** is used for secure communication
- Payment and personal data are encrypted during transmission
- Stored sensitive information is encrypted in the database

Benefit:

Prevents data theft during transmission and storage.



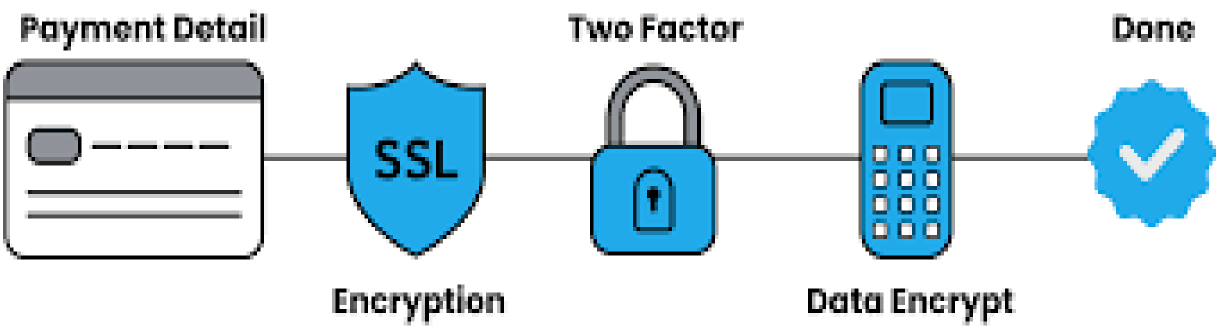
4. Secure Payment Integration

The system integrates with trusted payment gateways.

- Supports secure payment methods (UPI, cards, net banking)
- Payment processing handled by certified providers
- No sensitive card details stored in the system

Benefit:

Ensures safe and reliable financial transactions.



5. Input Validation and Sanitization

All user inputs are validated before processing.

- Prevents:
    - SQL Injection
    - Cross-Site Scripting (XSS)
    - Malicious data entry

Techniques:

- Server-side validation
- Client-side validation

Benefit:

Protects the system from common web attacks.

6. Session Management

The system securely manages user sessions.

- Session timeout after inactivity
- Secure session IDs
- Automatic logout for inactive users

Benefit:

Prevents session hijacking and unauthorized access.

7. Data Backup and Recovery

Regular backups are maintained to prevent data loss.

- Automated daily backups
- Secure storage of backup files
- Recovery mechanism in case of system failure

Benefit:

Ensures business continuity and data safety.

8. Audit Logs and Monitoring

The system maintains logs of user activities.

- Tracks:
    - Login attempts
    - Booking changes
    - Payment updates

Benefit:

Helps in detecting suspicious activities and ensures accountability.

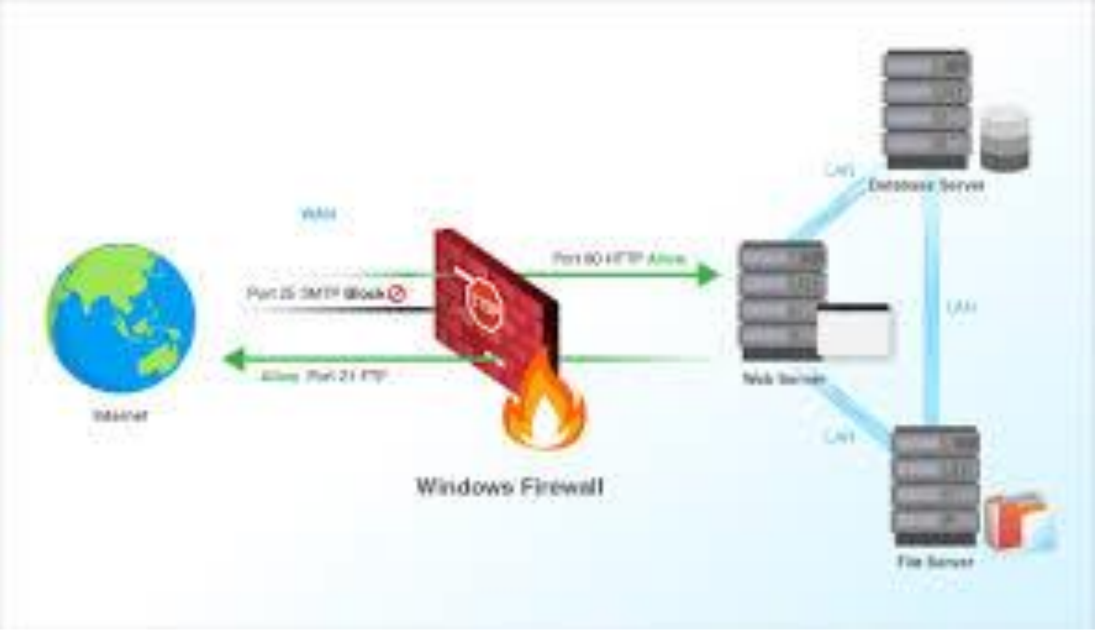
9. Firewall and Server Security

The application is protected at the server level.

- Firewall blocks unauthorized access
- Secure server configurations
- Protection against DDoS attacks

Benefit:

Enhances overall system security.



10. Privacy Protection

Customer data privacy is maintained.

- Data is not shared without consent
- Compliance with data protection policies
- Secure storage of personal information

Benefit:

Builds trust and ensures legal compliance.

Advantages

The CRM application for banquet hall booking offers numerous advantages by automating processes, improving customer interaction, and ensuring efficient management of bookings and events. It helps both business owners and customers by providing a smooth and organized system.

1. Improved Customer Management

The CRM system stores complete customer details such as contact information, booking history, and preferences in a centralized database. This enables staff to quickly access customer data and provide personalized services.

Benefit:

Enhances customer satisfaction and builds long-term relationships.

2. Efficient Booking Process

The application automates the entire booking process from inquiry to confirmation. It reduces manual work and ensures accurate scheduling of banquet halls.

Benefit:

Prevents double bookings and saves time.

3. Centralized Data Storage

All information related to customers, bookings, events, payments, and feedback is stored in one system.

Benefit:

Easy access, better organization, and reduced data redundancy.

4. Better Communication

The system enables seamless communication between customers and staff through automated emails, SMS notifications, and reminders.

Benefit:

Ensures timely updates and reduces miscommunication.

5. Time and Cost Efficiency

Automation of tasks such as booking management, billing, and reporting reduces the need for manual intervention.

Benefit:

Saves operational time and reduces labour costs.

6. Accurate Billing and Payment Tracking

The CRM system generates invoices and tracks payments efficiently.

Benefit:

Minimizes financial errors and ensures transparency in transactions.

7. Enhanced Decision Making

The system provides reports and analytics such as booking trends, revenue, and customer behaviour.

Benefit:

Helps management make informed business decisions.

8. Task and Event Management

The application helps in organizing event-related tasks such as catering, decoration, and scheduling.

Benefit:

Ensures smooth event execution without confusion.

9. Data Security and Reliability

The system includes security features such as authentication, encryption, and backups.

Benefit:

Protects sensitive customer and financial data.

10. Scalability

The CRM system can be expanded as the business grows.

Benefit:

Supports increasing number of customers and bookings without performance issues.

<u>Limitations</u>

Although the CRM application for banquet hall booking provides many advantages, it also has certain limitations that must be considered. Understanding these limitations helps in improving the system in future versions.

1. Initial Development Cost

Developing a CRM system requires investment in technology, software tools, and skilled developers.

Limitation:

Small businesses may find it expensive to implement initially.

2. Training Requirement

Staff members need proper training to use the system effectively.

Limitation:

- Time is required for learning
- Resistance from non-technical staff

3. Dependency on Internet and System Availability

The application usually depends on internet connectivity and server uptime.

Limitation:

- System cannot be accessed during network issues
- Server downtime may affect operations

4. Data Security Risks

Even though security measures are implemented, there is always a risk of cyber threats.

Limitation:

- Possible data breaches if security is weak
- Requires continuous monitoring and updates

5. Maintenance and Updates

The system requires regular maintenance and updates to function efficiently.

Limitation:

- Additional cost for maintenance
- Requires technical support

6. Limited Customization (in basic versions)

Basic CRM systems may not fully match the specific needs of every banquet hall business.

Limitation:

- Custom features may require additional development
- May not support all unique business workflows

7. Integration Challenges

Integrating the CRM system with third-party services like payment gateways or SMS APIs can be complex.

Limitation:

- Compatibility issues
- Additional configuration effort

8. Data Entry Errors

Incorrect data entered by users can affect the system’s accuracy.

Limitation:

- Wrong booking details
- Miscommunication due to incorrect inputs

9. Over-Reliance on Automation

Excessive dependence on the system may reduce human involvement.

Limitation:

- Lack of personal touch in customer interaction
- Issues if system fails suddenly

10. Scalability Challenges (if poorly designed)

If the system is not designed properly, it may face performance issues as the number of users increases.

Limitation:

- Slow performance
- System crashes under heavy load

Future Enhancements

The CRM application for banquet hall booking can be further improved by integrating advanced technologies and adding new features to enhance user experience, efficiency, and scalability. Future enhancements will help the system adapt to changing business needs and technological advancements.

1. Mobile Application Development

A dedicated mobile app can be developed for both customers and management.

Enhancement:

- Easy booking through smartphones
- Real-time notifications and updates

Benefit:

Improves accessibility and user convenience.

2. Online Booking with Real-Time Availability

Integrating a live availability system will allow customers to check hall availability instantly.

Enhancement:

- Calendar-based booking system
- Instant booking confirmation

Benefit:

Reduces manual interaction and speeds up the booking process.

3. AI-Based Chatbot Support

An AI-powered chatbot can be integrated to handle customer queries.

Enhancement:

- 24/7 customer support
- Automated responses for FAQs

Benefit:

Improves customer engagement and reduces workload on staff.

4. Advanced Analytics and Reporting

The system can include advanced analytics features using data visualization.

Enhancement:

- Revenue forecasting
- Customer behaviour analysis
- Peak booking time identification

Benefit:

Helps in better decision-making and business planning.

5. Integration with Third-Party Services

Future versions can support seamless integration with external platforms.

Enhancement:

- Payment gateways
- SMS and email services
- Event vendors (catering, decoration)

Benefit:

Creates a complete ecosystem for event management.

6. Cloud-Based Deployment

Migrating the system to cloud platforms can improve scalability and performance.

Enhancement:

- Use of cloud services like AWS or Azure
- Remote access from anywhere

Benefit:

Ensures high availability and data security.

7. Multi-Language Support

Adding multiple language options can make the system accessible to a wider audience.

Enhancement:

- Regional language support
- User-friendly interface

Benefit:

Enhances usability for diverse users.

8. Automated Reminder System

An advanced reminder system can be implemented.

Enhancement:

- Event reminders
- Payment due notifications

Benefit:

Reduces missed deadlines and improves communication.

9. Customer Loyalty Program

Introduce reward systems for repeat customers.

Enhancement:

- Discount offers
- Membership benefits

Benefit:

Encourages customer retention and loyalty.

10. Enhanced Security Features

Future updates can include stronger security mechanisms.

Enhancement:

- Biometric authentication
- Advanced encryption methods

Benefit:

Ensures higher data protection and trust.

Conclusion

The CRM application for banquet hall booking provides a comprehensive and efficient solution for managing customer interactions, reservations, event planning, and financial transactions. By integrating various modules such as lead management, booking confirmation, event coordination, billing, and feedback collection, the system ensures a smooth and organized workflow.

The use of structured designs like ER diagrams and DFDs helps in clearly understanding both the data relationships and process flow within the system. This not only improves system design but also enhances implementation efficiency. The application reduces manual effort, minimizes errors, and ensures better utilization of resources.

Additionally, the CRM system improves customer satisfaction by providing timely communication, personalized services, and a seamless booking experience. Features like automated notifications, centralized data storage, and secure payment handling contribute to a reliable and user-friendly system.

Despite certain limitations such as initial cost and maintenance requirements, the benefits of the system outweigh the challenges. With future enhancements like mobile applications, AI integration, and cloud deployment, the system can be further improved to meet modern business demands.

In conclusion, the CRM application serves as a powerful tool for banquet hall management, helping businesses operate more efficiently, enhance customer relationships, and achieve long-term growth and success.